* Capgemini – Accenture – Wipro – TCS – Cognizant :→ { Question }

→ Implement a Stack (LIFO) using Queue (FIFO)

Step	Operation	q1	q2	output
1	Start			
2	✓ push(10) → add to q2	[ ]	[10]	
3	swap	[10]	[ ]	
4	✓ push(20) → add to q2	[10]	[20]	
5	move q_1 to q_2	[ ]	[20 10]	
6	swap	[20 10]	[ ]	
7	✓ push(30) → add to q2	[20 10]	[30]	
8	move q1 to q2	[ ]	[30 20 10]	
9	swap	[30 20 10]	[ ]	
10	pop	[20 10]		[30]

Example data :→

✓ push (10);
✓ push (20);
✓ push (30);

top → 30, 20, 10

pop ();
→ 30

{ (i) Put new elements to q2
(ii) Move all elements q1 to q2
(iii) Swap q1 & q2

* Queue Using Stack → { Oracle }

DRY RUN TABLE :→

Step	Operation	S1	S2	output
1	start	[ ]	[ ]	—
2	enqueue(10)	[10]	[ ]	
3	enqueue(20)	[10 20]	[ ]	
4	enqueue(30)	[10 20 30]	[ ]	
5	dequeue → move S1 → S2	[10 20]	[30]	
6	move S1 → S2	[10]	[30 20]	
7	move S1 → S2	[ ]	[30 20 10]	
8	pop from S2	[ ]	[30 20]	10

enqueue (10);
enqueue (20);
enqueue (30);
dequeue ();
→ 10 (O/P)

* S1 → used for enqueue
* S2 → used for dequeue

* When dequeue happens and S2 is empty → move all elements from S1 to S2

Yesterday : (5:45 to 7:00 PM) ⇒ HCL (BNMIT / PESIT)
(ECE / EEE) ⇒ (8 – 10 LPA) (Time Duration) (5 mins)

* For a integer value N, generate binary numbers from 1 to N in this exact given format.

{ No data type validation }

Input ⇒ N = 5
Output ⇒ 1 10 11 100 101

queue<String> q = ["1"]; Concatenation:

Step 1: Pop "1" Print "1" Push "10" & "11" ["10" "11"]
Step 2: Pop "10" Print "10" Push "100" & "101" ["11" "100" "101"]
Step 3: Pop "11" Print "11" Push "110" & "111" ["100" "101" "110" "111"]
Step 4: Pop "100" Print "100" Push "1000" & "1001" ["101" "110" "111" "1000" "1001"]
Step 5: Pop "101" Print "101" Push "1010" & "1011" ["110" "111" "1000" "1001" "1010" "1011"]

{ 1 10 11 100 101 } Output